

Prolactin

Test Details

METHODOLOGY

Electrochemiluminescence immunoassay (ECLIA)

RESULT TURNAROUND TIME

Within 1 day

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Use

First test for work-up of galactorrhea (inappropriate lactation). Pituitary function test useful in the detection of prolactin-secreting pituitary tumors (microadenomas, macroadenomas) with or without galactorrhea, with or without structural evidence of sellar enlargement. An adult female premenopausal patient having amenorrhea and galactorrhea is highly suspect of pituitary prolactinoma and is a candidate for radiologic evaluation of the pituitary as well as serum prolactin levels. Elevated prolactin may be associated with corpus luteum insufficiency or anovulation. Sequelae of hyperprolactinemia include amenorrhea, anovulation, and decreased bone density.

Limitations

When determining prolactin, it should be remembered that the measured concentration is dependent upon when the blood sample was taken, since the secretion of prolactin occurs in episodes and is also subject to a 24-hour cycle.

The release of prolactin is promoted physiologically by suckling and stress. In addition, elevated serum prolactin concentrations are caused by a number of pharmaceuticals (e.g., dibenzodiazepines, phenothiazine), TRH, and estrogen.¹⁻³ The release of prolactin is inhibited by dopamine, L-dopa, and ergotamine derivatives.

In patients receiving therapy with high biotin doses (i.e., >5 mg/day), no sample should be taken until at least eight hours after the last biotin administration.⁴ As with all tests containing monoclonal mouse antibodies, erroneous findings may be obtained from samples taken from patients who have been treated with monoclonal mouse antibodies or who have received them for diagnostic purposes.⁴ In rare cases, interference due to extremely high titers of antibodies to ruthenium can occur.⁴ Extremely high titers of antibodies to streptavidin can occur in isolated cases and cause interference.⁴

Custom Additional Information

Levels rise during pregnancy and are elevated during lactation, in postpartum subjects, and following bilateral oophorectomy. Destructive pituitary diseases cause low levels. Hypothalamic lesions may be associated with increased values. Many pituitary tumors which previously were called chromophobe adenomas are now recognizable as prolactinomas.

Patients with hyperprolactinemia may have the multiple endocrine neoplasia syndrome, MEN-1.⁶ Provocative tests used in work-up of hyperprolactinemia include metyrapone stimulation of ACTH-2 and TRH provocative test.⁷

Antipsychotic drugs may elevate serum prolactin. Antipsychotics block dopamine, thereby elevating serum prolactin levels. Hyperprolactinemia is present in many patients receiving neuroleptics with an occasional patient developing amenorrhea, galactorrhea, and/or decreased libido. Amoxapine, a dibenzoxazepine type of tricyclic with antidepressant and antipsychotic characteristics, has been found to cause galactorrhea and oligomenorrhea with hyperprolactinemia. Amoxapine may have a dopamine blocking action.⁸ The prolactin level may rise significantly but only briefly. Point prolactin level determinations during therapy may be within normal range while total integrated 24-hour secretion is significantly increased. It has been recommended that patients who develop amenorrhea and/or galactorrhea during neuroleptic therapy should be observed regularly for possible emergence of a pituitary tumor.

Persistent elevations of plasma prolactin levels may be observed with, and after withdrawal from, chronic cocaine abuse, and may reflect a cocaine-induced derangement in the neural dopaminergic regulatory systems.⁹

Specimen Requirements

Specimen

Serum (preferred) **or** plasma

Volume

1 mL

Minimum Volume

0.7 mL (**Note:** This volume does **not** allow for repeat testing.)

Container

Red-top tube, gel-barrier tube, **or** green-top (lithium heparin) tube. Do **not** use oxalate, EDTA, or citrate plasma.

Collection Instructions

If a red-top tube or plasma is used, transfer separated serum or plasma to a plastic transport tube.

Stability Requirements

Temperature	Period
Room temperature	7 days
Refrigerated	14 days
Frozen	14 days
Freeze/thaw cycles	Stable x3

Reference Range

See table.⁵

Age ^{1,5}	Male (ng/mL)	Female (ng/mL)
0 to 30 d	4.3–32.8	4.5–24.4
1 to 6 m	5.4–51.4	5.4–51.4
7 m to 1 y	5.3–28.9	4.8–33.4
2 to 12 y	1.8–44.2	4.8–33.4
13 to 30 y	3.6–31.5	4.8–33.4
31 to 50 y	3.9–22.7	4.8–33.4
51 to 80 y	3.6–25.2	3.6–25.2
>80 y	3.6–32.0	3.6–32.0

Storage Instructions

Refrigerate.

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Footnotes

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